

Four Levers, Four Corners: Attention-Sink Signatures Dissociate in From-Scratch Vision–Language Pretraining — A Precondition for Grounding Fidelity

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Abstract. Trained language models show three effects at the same early token. Attention concentrates on the token. The value vector at that token has a very small norm. The residual stream at that token has a very large norm. These three effects occur together so often that the literature treats them as one “attention sink” phenomenon. We test that reading in a setting where a reader can watch the three effects form. That setting is from-scratch multimodal pretraining. We train a 222M-parameter vision–language model. A SigLIP-B/16 encoder feeds a randomly initialized decoder with the SmoLLM2-135M architecture. We train this model under four levers: standard softmax attention, output-gated softmax, unnormalized sigmoid attention, and decoder initialization from a pretrained text language model. We log all three signatures as separate per-head quantities through training. The three signatures come apart. Across three seeds, the four levers reach four different corners of (concentration $\times$ value-norm $\times$ massive-activation) space. No two arms share a signature triple. The value-norm ratio alone moves in three directions. The lever drains it (0.38–0.72), leaves it near 1, or amplifies it (1.48–1.60). We then train one repetition-confound-free run over one billion fresh tokens (single seed). In that run massive activation grows by 130 percent while attention concentration stays at exactly zero. The per-head correlation between concentration and value-norm also flips sign across arms (+0.67 to -0.76 , pooled -0.20). No fixed-sign coupling links the two axes. Earlier text-only work separates massive activations from attention sinks. We show, for the first time in from-scratch multimodal pretraining, that all three signatures respond independently to ordinary training-time choices. Value-norm drain is one of the three. This result speaks to grounding as a precondition question. Attention mass that a positional artifact captures is attention that the model does not spend on image evidence. Other work links sink-like attention to hallucination in deployed vision–language models. We do not test whether signature dissociation predicts grounding behavior. We establish when and how each signature forms, which is the step that comes first.

1. Introduction

Decoder-only transformers learn a habit early in training. They send a large share of attention to the first token or tokens of a sequence, whatever those tokens contain. The habit has practical weight. Streaming inference methods depend on it [12]. The extreme activation outliers that come with it make quantization harder [10]. A survey now organizes a subfield around how to interpret the habit and how to remove it [13].

In text language models the sink does not arrive alone. Three measurements move together. Attention concentrates on the sink token. The value vector at that token drops to a near-zero norm, which the literature calls value-state drain [6]. The residual stream at that token grows an abnormally large norm, which the literature calls a massive activation [10, 11]. Gu et al. [1] report that these effects settle on the same few tokens. Queipo-de-Llano, Arroyo et al. [2] report that they appear early in training, near step 1000. This regular co-occurrence has led the field to treat the three signatures as facets of one attention-sink phenomenon.

The field disputes whether they really are one phenomenon. One line of work argues for causal unity. Massive activations in the residual stream mathematically require representational compression. Ablation of those acti-

vations removes both compression valleys and sink formation [2]. A related view holds that outlier-driven rescaling by attention and residual sinks is *essential* to stable transformer training [14]. Two recent text-only studies pull the other way. Each separates a pair of these signatures by intervention. A change of normalization scheme crushes the massive-activation spike, and the sink ratio survives [3]. A value-scale intervention in from-scratch text language models keeps the sink and suppresses massive activations [4]. Both results are text-only. Each one separates at most two of the three signatures.

Vision-language models make a sharp instrument for this question, for two reasons. First, the multimodal setting changes what position 0 is. Our sequence starts with 49 image tokens and has no BOS token. The candidate sink token is therefore a visual token. It has none of the special-token machinery that text-LM sink accounts use. Second, the multimodal sink studies we know analyze frozen, already-trained backbones at inference time [7, 8]. They show that vision-side and language-side sinks have different origins. A frozen model cannot answer a question about emergence. A reader can only see whether the three signatures arrive together, in sequence, or independently while the signatures form. That requires from-scratch pretraining with all three logged separately from step 0. To our knowledge no prior work does this in a multimodal model.

The question also carries deployment stakes. Attention allocation is the mechanism that connects the language a vision-language model generates to the content of the image. A sink is a measurable failure mode of *where* that allocation goes. A growing literature ties these same signatures to hallucination in deployed vision-language models. Visual attention sinks absorb attention, massive activation of specific hidden dimensions drives them, and redistribution methods recover the absorbed attention [21]. Hallucination errors cluster in the decoding steps immediately after the model generates a sink token, which motivates sink-aware decoding [22]. One paper proposes a causal chain from positional encoding through massive activations to visual sinks and then to hallucination [23]. In text language models, sink attention together with value-norm structure carries enough signal to detect hallucinations [24]. All of that work probes or intervenes on models that are already trained. The question underneath it is when these signatures form during training, and whether they form as one thing or as several. This paper answers that question. We do not measure whether their dissociation predicts grounding or hallucination behavior. This paper establishes the training-dynamics precondition.

We work at deliberately small scale. The model is a 222M-parameter nanoVLM [18]. A pretrained SigLIP-B/16 encoder [16] feeds a randomly initialized decoder with the SmolLM2-135M architecture [17]. We train it under four levers. Each lever targets one sink-relevant mechanism. *baseline* uses standard softmax attention. *g1-gate* adds a zero-initialized elementwise output gate on attention, which removes both the sink and massive activations in text language models [5]. *sigmoid* replaces softmax with unnormalized sigmoid attention. That removes the normalization from which sinks are argued to come [1]. *textinit* initializes the decoder from the pretrained SmolLM2 text language model, and thus imports whatever sink structure text pretraining built. A validated probe logs concentration, value-norm ratio, and massive activation for each (layer, head) pair every 100 steps. The four-arm comparison reuses a small image pool at high epoch counts. We therefore re-test the central negative result on a fresh, non-repeated stream of one billion tokens (*RF*). That run removes the overfitting confound.

Contributions.

1. **Four levers, four corners (n = 2–3 seeds/arm).** The four arms reach four different corners of (concentration $\times$ value-norm $\times$ massive-activation) space. No two arms share a signature triple. The value-norm ratio alone moves in three qualitatively different directions, drained or unchanged or amplified, and the lever decides which (Fig. 1, Table 2).

2. **Repetition-confound-free decoupling at 1B tokens ($n = 1$).** On fresh, non-repeated data, massive activation rises 130% (h-ratio $1.43 \rightarrow 3.22$) across a full billion tokens. Attention concentration stays at exactly zero for the entire single-seed run. No head ever crosses the sink threshold (§3.2, §5).
3. **No fixed-sign head-level coupling.** The per-head correlation between concentration and value-norm flips sign across arms ($+0.67$ baseline $\rightarrow -0.76$ *textinit*, pooled -0.20). No fixed-sign coupling links the two axes (§3.3).

Text-only work already separates massive activations from concentration through normalization [3] and value-path [4] interventions. Decoupling by itself is therefore not new, and we do not claim it. The new part is the conjunction. All three signatures respond separately to ordinary levers during from-scratch *multimodal* pre-training, and value-norm drain is a third axis that moves on its own. No prior decoupling study and no prior co-emergence study has examined this setting.

2. Setup

Model and token layout. All runs use a 222M-parameter nanoVLM [18]. A pretrained, trainable SigLIP-B/16 vision encoder [16] feeds a decoder with the SmolLM2-135M architecture [17] through a learned modality projector. The decoder has 30 layers of grouped-query attention. Each layer has 9 query heads that share 3 KV heads. The decoder therefore has 270 (layer, query-head) pairs but only 90 (layer, KV-group) value projections. This distinction matters when we count per-head observations (§3.3). The decoder trains **from random initialization** in every arm except *textinit*, because the point is to watch the signatures form. Each sequence holds 49 image tokens as a causal prefix, then 79 left-padded text tokens, for 128 tokens in all. **Position 0 is the first image token, and there is no BOS token.** What happens at position 0 is therefore a property of the visual prefix. It is not inherited BOS machinery.

Arms. We use four training levers. Each lever targets one sink-relevant mechanism. Everything else stays byte-identical across arms.

arm	attention	LM init	ViT init	lever precedent
<i>baseline</i>	softmax	random	pretrained	—
<i>g1gate</i>	softmax + elementwise σ -gate (zero-init, post-SDPA)	random	pretrained	G1 gating [5]
<i>sigmoid</i>	unnormalized sigmoid, no softmax	random	pretrained	Gu et al. [1]
<i>textinit</i>	softmax	pretrained SmolLM2-135M	pretrained	— (novel control)

The *g1gate* lever and the *sigmoid* lever have established sink effects in text-only models [5, 1]. The *textinit* lever has no precedent in the sink literature. It works as an inheritance control. It imports whatever sink structure text pretraining already built into SmolLM2.

Data and the two training regimes. The four-arm comparison trains on four curated subsets of the *cauldron* [19], which hold about 146K images. The arms are matched at about 100M tokens each. *textinit* stops at 60M tokens, because it reaches its three-signature floor earlier (§3.1). Reuse of a 146K-image pool gives high visual-epoch counts. A reader could therefore treat a “no sink emerges” result as an overfitting artifact. The **RF** arm (random-fresh) answers that objection. It re-trains the *baseline* recipe on a fresh FineVision stream [20] to 1B tokens. That stream holds about 4.6M natural images and gives 2.39 effective visual epochs, against about

74 epochs for the repeated pool. A change of dataset trades the repetition confound for a domain-shift confound. We accept that trade and document it. The fresh pool holds natural images, leans heavily on COCO, and overlaps the repeated subsets by less than 3%. We did not run a third, domain-matched control (§5).

Three signatures, tracked separately. We log the three sink symptoms that the text-LM literature reports together as independent per-(layer, head) quantities. The metric conventions follow Gu et al. [1] at a fixed sequence length.

- **Concentration** — $\text{Sink}_1^\varepsilon$: the fraction of (layer, head) pairs whose *mean* attention to position 0 is above ε . We also log the mean and the maximum attention to position 0. We use the $\varepsilon = 0.3$ default of [1] and check robustness at $\varepsilon \in \{0.2, 0.4\}$. Cross-arm tables report the stricter $\varepsilon = 0.2$, which makes an absence claim harder to pass.
- **Value-norm ratio** (v-ratio) — $\|\mathbf{v}\|$ at position 0 divided by the mean $\|\mathbf{v}\|$ of the other positions. A value below 1 is value-drain [6]. A value above 1 is amplification. Under grouped-query attention the value vector belongs to a KV group. Each per-(layer, query-head) v-ratio therefore repeats its group value across 3 query heads, which gives 90 independent value observations, not 270.
- **Massive activation** (h-ratio) — the residual-stream norm at position 0 divided by the mean norm of the other positions.

All three metrics **anchor on position 0 by construction**. We state this as a measurement choice, and we check it. At seed 0, per-position attention mass makes position 0 the maximum-mass token in every arm (Appendix C). Section 5 handles the remaining seed-level caveat. The *sigmoid* arm reports the row-normalized attention view, which keeps concentration comparable across arms. We also log the raw sigmoid mass.

Probe. The function `probe_sinks()` re-walks the decoder from the live module weights in eager mode, in `fp32`, with no gradients. It is independent of the training path, which uses fused SDPA kernels, autocast, and `torch.compile`. Every call validates its hidden states against the real forward pass of the model, to a relative error below 10^{-2} . The probe therefore cannot drift from what the trained model computes. The probe batch stays fixed across all runs and all seeds, so every number in this paper is comparable from run to run. Probes fire every 100 optimizer steps. That cadence is dense enough to timestamp the first threshold crossing of each signature (§3.4).

Recipe. We use AdamW with weight decay 0.1, following [1], and a gradient clip of 1.0. The schedule is cosine with 3% warmup. The learning rates are $4\text{e-}4$ for the language model, $2\text{e-}3$ for the projector, and $1\text{e-}4$ for the vision encoder. We use `bf16` autocast, `torch.compile`, and a batch size of 128. Arms in a comparison differ only in the lever under test. Training stays healthy in all reported runs. On the 1B fresh run, held-out fresh validation loss falls from 1.46 to 0.638 and tracks train loss with no overfit gap.

3. Results

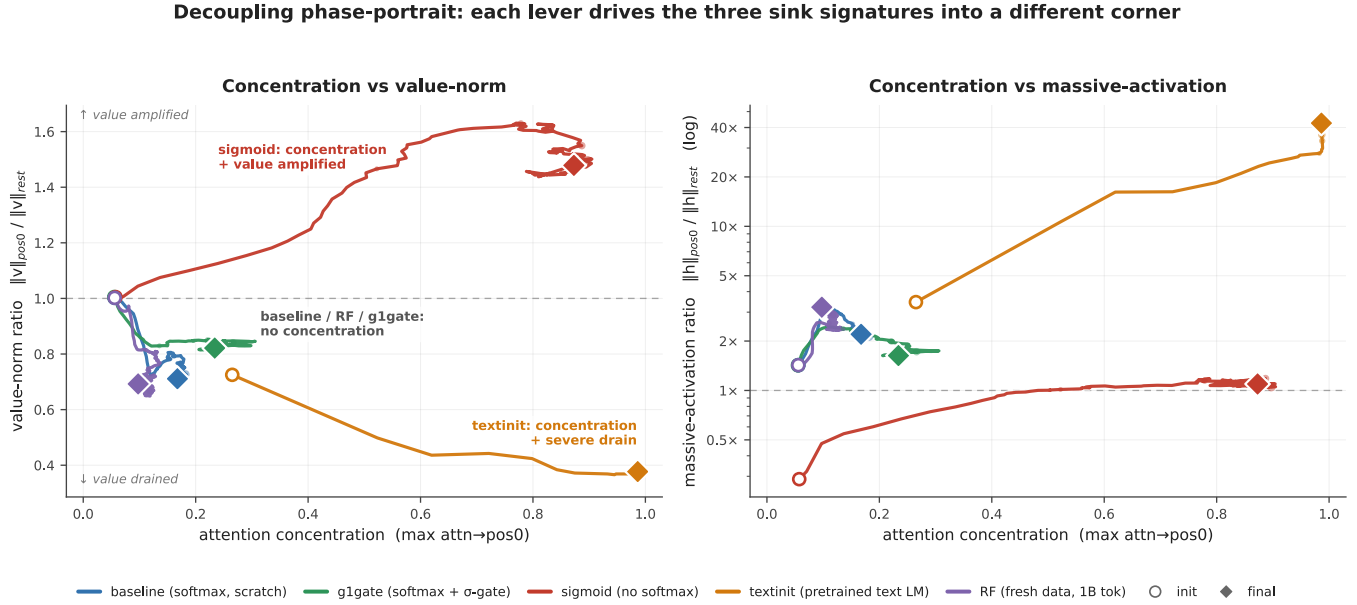


Figure 1: Decoupling phase-portrait. Training trajectories of every arm in (concentration $\times$ value-norm) space, at left, and in (concentration $\times$ massive-activation) space, at right, on a log scale. Circles mark initialization. Diamonds mark the final checkpoint. Each lever drives the signatures to a different corner. *sigmoid* (red) reaches strong concentration, *amplifies* the value norms, and shows no massive activation. *textinit* (orange) combines total concentration with severe value-drain and an extreme massive activation. *baseline*, *g1gate*, and *RF* (blue, green, violet) never leave the no-concentration region, and their massive activation grows. If the three signatures were one phenomenon, these trajectories would move together along one direction. Instead every combination of directions occurs.

3.1 Four training levers reach four different signature corners

Table 1 reports the three signatures for all four arms and all seeds. *baseline* and *sigmoid* have two seeds each. *g1gate* and *textinit* have three. Each row is read at its final matched checkpoint: about 100M tokens for baseline, *g1gate*, and *sigmoid*, and 60M tokens for *textinit*. All three signatures of *textinit* have plateaued at 60M tokens. Its h-ratio changes by less than 10% from 60M to 100M in the seeds that continued. We trust the seed-0 values from a checksummed archive rather than re-derive them first-hand (§5).

Table 1 — signature triple by arm, all seeds. Sink^{0.2} is the fraction of the model’s 270 (layer, head) pairs whose mean attention→pos0 exceeds 0.2.

arm	seed	tokens	Sink ^{0.2}	mean attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	0.723	2.16
baseline	s1	100M	0.000	0.062	0.687	1.71
g1gate	s0	101M	0.004	0.068	0.805	1.67
g1gate	s1	100M	0.011	0.073	0.845	2.04
g1gate	s2	100M	0.0037	0.072	0.854	2.24
sigmoid	s0	102M	0.830	0.377	1.479	1.10
sigmoid	s1	100M	0.756	0.311	1.603	1.30
textinit	s0	60M	0.852	0.627	0.377	42.5
textinit	s1	60M	0.556	0.235	0.634	5.50
textinit	s2	60M	0.578	0.232	0.484	12.2

Table 2 collapses the seeds into per-arm ranges. **No two arms share a signature triple.** The value-norm axis alone takes three qualitatively different directions. The lever drains it below 1, leaves it near 1, or amplifies it above 1.

Table 2 — the four corners.

arm	concentration	value-norm	massive activation
baseline	absent (0.000)	mild drain (0.69–0.72)	moderate (1.7–2.2)
g1gate	suppressed (0.004–0.011)	neutral (0.81–0.85)	moderate (1.7–2.2)
sigmoid	strong (0.76–0.83)	amplified (1.48–1.60)	none (1.1–1.3)
textinit	total (0.56–0.85)	strong drain (0.38–0.63)	extreme (5.5–42.5)

The corners separate pairwise on single axes. *g1gate* differs from *baseline* on the value-norm axis alone. Concentration is absent or nearly absent in both arms, and massive activation is moderate in both. The gate nonetheless neutralizes the mild value-drain that the baseline develops. *sigmoid* and *textinit* both reach strong concentration, and they then part on two axes. They differ in the *direction* of the value-norm move, amplified against drained, and they differ in massive activation, none against extreme. One lever moves one axis and leaves the other axes where a different lever put them. Figure 1 shows the four trajectories fanning out from a common origin.

Reproducibility differs by arm. *g1gate* is the tightest arm. Its Sink^{0.2} values are 0.004, 0.011, and 0.0037 across three seeds, which is at most 1% of heads. *baseline* and *sigmoid* are tight in both of their seeds. The corner of *textinit* repeats across seeds. Its magnitudes do not. All three *textinit* seeds show high concentration, strong value-drain, and an h-ratio above 5. Seed 0 nonetheless sits far above the other two seeds on every signature at once. Sink reads 0.85 against 0.56–0.58, mean attn→pos0 reads 0.63 against 0.23, and h-ratio reads 42.5 against 5.5–12.2. The h-ratio has plateaued by 60–100M tokens in both lower seeds. The spread is therefore genuine seed sensitivity, not an unconverged transient. We report the massive activation of *textinit* as a **range (5.5–42.5×) with a median near 12×** everywhere in this paper. We treat the corner as the reproducible claim, and no particular magnitude.

3.2 Massive activation grows across 1B fresh tokens while concentration stays at zero

The four-arm comparison reuses a 146K-image pool at high epoch counts. A reader could therefore read its “concentration never emerges” result as an overfitting artifact. The RF arm removes that confound. It runs the identical *baseline* recipe on a fresh, non-repeated FineVision stream to one billion tokens, which is 2.39 effective visual epochs.

Concentration never arrives. **Sink^{0.3} = 0.000 across the entire 0 → 1B run.** No head of the 270 crosses the sink threshold, at any of about 700 probes. Massive activation, in the same run, keeps growing far past warmup.

Table 3 — RF (fresh data, seed 0), init → 1B tokens.

signature	@ init	@ 1B	net
Sink ^{0.3} (concentration)	0.000	0.000	flat zero, entire run
max attn→pos0	0.056	0.098	≈flat, far below threshold
h-ratio (massive activation)	1.43	3.22	+130% , still rising at 1B
v-ratio (value-norm)	1.00	0.69	net drain, non-monotone

Warmup does not explain the rise in h-ratio. The h-ratio climbs monotonically from 2.40 at about 57M tokens to 3.22 at 1B tokens, long after the 3% warmup window closes. In Figure 1, at right, the violet trajectory climbs and never moves rightward. The v-ratio ends below its value at initialization. About 75% of that drop happens in the first 57M tokens, and the v-ratio then recovers part of it. We therefore report the v-ratio as supporting context, not as ongoing emergence. **Massive activation grows across a full billion tokens of multi-modal training while attention concentration never leaves zero.** This run uses a single seed (§5).

3.3 No universal head-level coupling between concentration and value-norm

Tight coupling between concentration and value-drain should at minimum hold the sign of their per-head correlation constant across training regimes. It does not. Table 4 reports the Pearson r between per-head attention→pos0 and value-norm ratio. We read it at the final checkpoint of each arm, over n = 270 (layer, query-head) pairs per arm. Appendix Fig. A2 plots the scatter.

Table 4 — per-head r(attn→pos0, v-ratio), final checkpoint. ***p < 0.001; ns = not significant. See the independence note below.

baseline	g1gate	sigmoid	textinit	RF	pooled (n = 1350)
+0.67***	+0.53***	−0.03 ns	−0.76***	+0.43***	−0.20***

Two honesty notes apply to the n. First, the decoder uses grouped-query attention, where 9 query heads share 3 KV heads per layer (§2). The 270 pairs per arm therefore hold only 90 independent value-norm observations, because each KV group shares its value vector across 3 query heads. The attention side is genuinely per-query-head. Second, the significance stars treat the pairs as independent, and heads within a layer and within a KV group are not independent. Read the stars as descriptive, not as inferential. Neither caveat touches the finding itself, which is about sign. Heads that attend more to position 0 have *larger* value norms in the baseline regime and *smaller* value norms under text initialization. The pooled correlation is weak only because arms of opposite sign cancel. A fixed-sign coupling would show the same sign everywhere. Each lever instead sets its own local relationship between the two axes.

3.4 Ordering: norm signatures lead, and concentration is late, mirrored, or absent

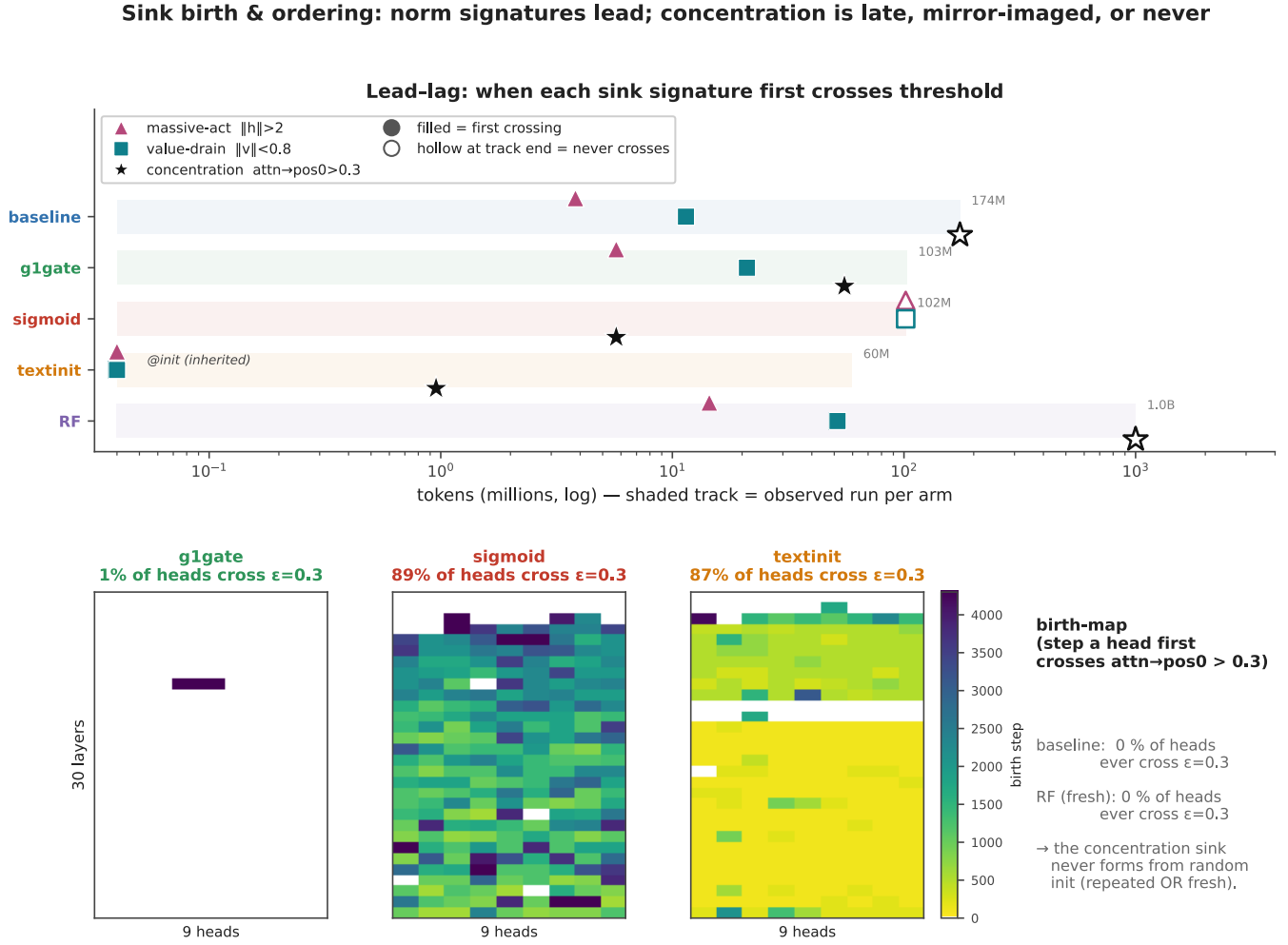


Figure 2: Sink birth and ordering. *Top:* a time-to-event view. Each shaded track spans the tokens for which we observed that arm, from 60M to 1B. A filled marker shows the first probe at which a signature crossed its per-head threshold: massive activation $h > 2$, value-drain $v < 0.8$, concentration $\text{attn} \rightarrow \text{pos0} > 0.3$. A hollow marker at the end of a track means that the signature never crossed inside the observed run. In the softmax-from-scratch arms, baseline and g1gate and RF, the norm signatures cross within about 4–50M tokens. Concentration in those arms never crosses, through 174M tokens for baseline and through 1B tokens for RF, or barely crosses, in g1gate. *sigmoid* mirrors that pattern. Its concentration crosses at about 6M tokens, and neither norm signature ever crosses. *textinit* inherits massive activation and value-drain at 0 tokens and crosses concentration before 1M tokens. *Bottom:* birth-maps, which show the step at which each (layer, head) pair first crosses the concentration threshold. No baseline head and no RF head ever crosses. 1% of g1gate heads cross, against 89% for sigmoid and 87% for textinit.

Dense probing lets us timestamp the arrival of each signature (Figure 2). The ordering stays consistent inside each arm family. In the softmax-from-scratch arms, *baseline* and *g1gate* and *RF*, the norm signatures come early. Massive activation crosses its per-head threshold within the first few to about 15M tokens, and value-drain follows within about 50M tokens. Concentration never comes. No baseline head and no RF head crosses the concentration threshold at any point in training. *g1gate* reaches 1% of heads, a single-layer blip late in training. *sigmoid* mirrors that pattern. Its concentration crosses early, at about 6M tokens, and 89% of its heads cross eventually, while neither norm signature ever crosses. *textinit* inherits its signatures rather than grows them. Value-drain is present at 0 tokens, imported with the pretrained text-LM weights, and concentration crosses before 1M tokens.

The sink stripe: every query attends to the first image token — absent in baseline, total in textinit, and already present at init (inherited)

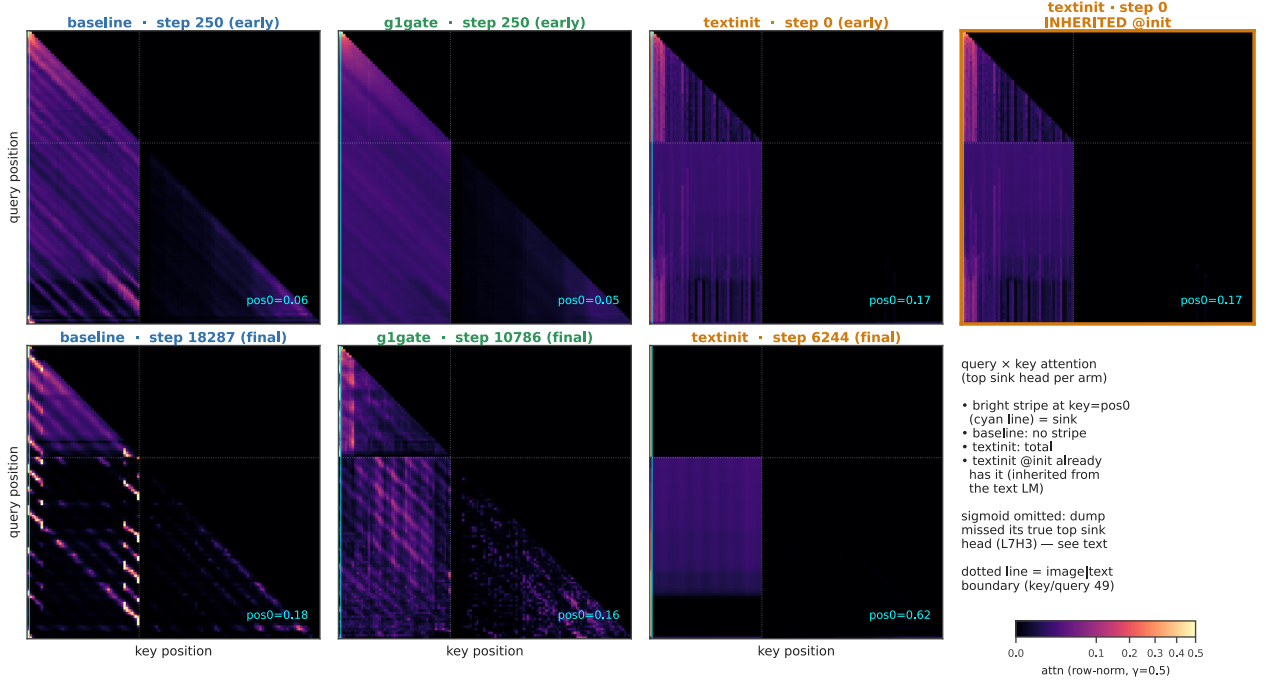


Figure 3: The sink stripe. Query $\times$ key attention of the top sink head of each arm, row-normalized. The top row is early in training. The bottom row is the final checkpoint. The cyan line marks key = pos0. The dotted line marks the image-to-text boundary. The pos0 stripe is *absent* in baseline throughout, and *total* in textinit, which reaches $\text{attn} \rightarrow \text{pos0} = 0.62$ at its final checkpoint. The rightmost panel shows the stripe *already present at step 0* in textinit, inherited from the text language model. We omit sigmoid. Its checkpoint dump selected heads by raw, unnormalized gate score and missed the true top sink head of the arm, L7H3, which reaches 0.87 normalized. A panel built from the dumped heads would therefore understate the concentration of sigmoid. Tables 1–2 and Figure 2 establish that concentration instead.

Figure 3 shows the same result at the level of a single head. It plots the query $\times$ key attention map of the top sink head of each arm. The vertical stripe at key = pos0 marks every query attending to the first image token. That stripe is absent in *baseline* at both the early and the final checkpoint, and total in *textinit*. The rightmost panel carries the most weight. **The stripe is already there at step 0 in textinit**, imported with the text-LM weights before the model has seen one image. We omit *sigmoid* from this figure for the measurement reason that the caption gives. Tables 1–2 and Figure 2 carry the concentration result for that arm. Attention-entropy collapse, which the text-LM literature uses as the usual correlate of concentration, separates the same way. Only *sigmoid* and *textinit* collapse (Appendix Fig. A3). Entropy collapse tracks the concentration axis, not the norm axes.

A positional footnote. In *textinit* the signatures partly decouple in *position* as well as in magnitude. In some seeds the most-attended token, the residual-norm peak, and the value-drain minimum sit on three different tokens. Attention sits at pos1, for example, while the norm peak and the drain sit at pos13. We report this as a caveat on where to anchor the magnitudes of textinit (§5), not as an independent finding.

4. Related Work

Attention sinks and their companions in text language models. Gu et al. [1] give the canonical empirical account. The sink token acts “more like key biases, storing extra attention scores, which could be non-informa-

tive and not contribute to the value computation.” They report small key and value norms at the sink as part of the same phenomenon, and they connect the sink to massive residual-stream activations [10, 11]. They also show that unnormalized sigmoid attention prevents sink formation in text language models up to 1B parameters. Our *sigmoid* arm builds on that result. Guo et al. [6] describe the coupling between concentration and value-drain as “active-dormant” heads, where the model actively drives the value state of the sink head toward zero. Queipo-de-Llano, Arroyo et al. [2] make the strongest unity claim, and the only genuinely *causal* one. Massive activations mathematically require representational compression, and ablation of the layer-0 massive activation of a model removes both compression valleys and sink formation. Across Pythia checkpoints they find that sinks, compression valleys, and massive activations appear together near step 1000 and then stay synchronized. We reserve the term “causally unified” for that result alone. Peng et al. [15] trace a first-position sink circuit that emerges early in from-scratch text pretraining. That work covers from-scratch emergence dynamics, in text only, and does not separate the signatures.

Prior decoupling results: text-only, two axes. Two recent papers already separate pairs of these signatures in text models, and we scope our claim around them. Sun, Canziani, LeCun and Zhu [3] show that massive activations and attention sinks are dissociable architectural artifacts. A change of normalization scheme crushes the massive-activation spike, and the sink ratio survives. That is a two-way dissociation, in text only, through a normalization lever, on trained checkpoints. Chen and Yao [4] decouple the same pair from the opposite direction and from scratch. In text language models of 0.1–0.3B parameters, probed at dense checkpoints, a value-scale intervention keeps the sinks and suppresses massive activations. Neither paper treats value-norm drain as a third axis that moves on its own. Neither paper is multimodal. On the opposite side of the argument, Qiu et al. [14] hold that outlier-driven rescaling by attention and residual sinks is essential to stable training. The field is unsettled on two counts. It disputes whether these signatures separate, and it disputes whether anyone should remove them at all.

Multimodal sinks: frozen backbones, inference time. Luo et al. [7] identify high-norm attention-sink tokens that originate in the vision transformer, and they separate ViT-propagated sinks from LLM-emerged sinks. Choi et al. [8] likewise distinguish vision-sinks from language-sinks in a frozen large vision–language model and gate them by layer. Both papers establish that multimodal sinks have distinct vision-side and language-side origins, and our *textinit* inheritance result fits that frame naturally. Both papers nonetheless analyze models that are already trained. Neither can observe emergence, and neither measures the three signatures as separate quantities. Vision transformers also grow high-norm “register” tokens of their own [9], which is why the pre-trained encoder gets its own limitation in §5.

Sinks and hallucination in deployed vision–language models. A parallel line of work ties these signatures to grounding failure. Kang et al. [21] attribute *visual* attention sinks to massive activation of specific hidden dimensions, and they redistribute the absorbed attention to reduce hallucination. Shukla and Kira [22] report that hallucination errors concentrate within a few decoding steps of the generation of a sink token. They then decode in a sink-aware way. Zhang et al. [23] trace a causal chain from rotary position encoding through massive activations to visual sinks and then to hallucination. In text language models, Binkowski et al. [24] detect hallucinations from sink structure. Their classifier relies preferentially on sinks whose value vectors have large norms, which makes it the closest precedent for treating value norms as a signal. All four papers work on models that are already trained, at inference time. Each one treats massive activation and concentration as one coupled mechanism, and [24] is text-only as well. None of them tracks the signatures across training. None of them treats value-norm drain as a third axis that moves on its own. Our result is the training-dynamics complement to that line. The coupling on which the line depends is lever-dependent, not fixed.

The gating lever. Qiu et al. [5] introduce the head-specific, zero-initialized elementwise sigmoid gate on attention output that our *g1gate* arm uses. In text language models the gate “largely reduces the attention score allocated to the first token and decreases massive activations” while it improves quality. Our finding sharpens what the gate does in the from-scratch multimodal setting. Concentration is already absent in the baseline. The isolated, reproducible effect of the gate is therefore to *neutralize value-drain*, from a baseline v-ratio of 0.69–0.72 to 0.81–0.85. The gate leaves massive activation untouched. That effect stays invisible unless the three signatures are logged separately.

Positioning. The closest prior work separates at most two of the three axes, in text-only models, through normalization or value-path interventions. The multimodal studies analyze frozen models. We are aware of no study that tracks concentration, value-norm, and massive activation as independently measured quantities across from-scratch multimodal pretraining. Our claim is therefore the conjunction: **first to show all three sink signatures independently controllable during from-scratch multimodal pretraining, adding value-norm drain as a third axis** beyond the massive-activation-vs-sink dissociations shown in text models [3, 4]. We do not claim priority on decoupling itself.

5. Limitations

Metrics anchored on position 0. We measure all three signatures at the first image token. We verified this anchoring first-hand at seed 0 through per-position attention *mass*, not through argmax alone. Position 0 is the maximum-mass token in every arm. The mean and max-head values are 0.06 and 0.17 for baseline, 0.07 and 0.23 for *g1gate*, and 0.30 and 0.66 for sigmoid. They are 0.63 and 0.99 for *textinit*, whose next-highest position reads 0.009 (Appendix C). That check validates the reported seed-0 magnitudes, which include the *textinit* h-ratio of 42.5. It does not close the question at seeds 1 and 2. Live-probe argmax data at those seeds shows that the most-attended position can migrate off position 0 in *textinit*. One seed splits across pos1 and pos13. Part of the cross-seed magnitude spread of *textinit* may therefore come from position migration rather than from magnitude noise alone. That is one more reason we report *textinit* as a range and a median, and treat its corner rather than its magnitudes as the claim. We leave a full per-position scan of value norms and residual norms across all seeds to a camera-ready revision.

Pretrained vision encoder. The SigLIP encoder is pretrained, so no arm is fully from scratch. Only the decoder is. Vision transformers grow high-norm register tokens of their own [9], and sinks can propagate from a vision transformer into a large vision–language model [7]. Part of our massive-activation signal could therefore be inherited rather than decoder-formed. Our defense is the trajectory. The h-ratio starts at 1.0–1.4 at initialization and *rises* through training, from 1.43 to 3.22 across 1B tokens in the RF arm. Pure inheritance from a static encoder predicts a high, flat h-ratio from step 0 instead. A control with a randomly initialized vision transformer, which would isolate the decoder entirely, is future work.

Token scale. Our runs reach at most 1B tokens per arm, against the roughly 5B tokens that are canonical in the text-LM sink literature [1]. Sink emergence is early relative to that budget. Text-LM sinks and their companions form near step 1000 [2], far inside our range. We nonetheless cannot rule out that a signature absent at 1B tokens emerges later. Confirmation at larger scale is future work.

Reproducibility of *textinit* magnitudes. The massive activation of *textinit* is seed-sensitive. The h-ratio spans 5.5–42.5 across three seeds, and seed 0 is the consistent outlier on every signature. The corner is the reproducible claim: high concentration, plus strong drain, plus large massive activation. No specific magnitude is.

Provenance and seed count. We trust the seed-0 raw probes for the four-arm comparison from a checksummed archive summary rather than re-derive them first-hand. We *did* independently re-derive seeds 1 and 2. That audit caught and corrected a metric-labeling error in an earlier internal consolidation, which mixed mean and max attention in one column. That correction is why we report the metrics like-for-like here. *baseline* and *sigmoid* have two seeds. *g1gate* and *textinit* have three. The RF arm uses a **single seed**, and it contains one weights-only resume at about 57M tokens that an out-of-memory error forced. The audit verified signature continuity across that seam. The v-ratio and h-ratio values are identical at the shared checkpoint. A double-covered 600-step overlap diverges only within probe noise, and concentration reads 0.000 on both sides. The decoupling movement also completes before the seam. Concentration was reproducibly zero across both seeds of the repeated-data baseline, which we take as adequate support for the negative claim. A second fresh-data seed would strengthen it.

Domain shift in the fresh-data control. The RF arm removes the repetition confound by a change of dataset, and drops the visual-epoch count from about 74 to 2.39. That change introduces a domain-shift confound in its place. We accepted the trade deliberately. Repetition is the dominant confound, and we chose the fresh pool to minimize shift, as natural images, COCO-heavy, with less than 3% overlap. The no-sink result also held in both the stronger-shuffle and the weaker-shuffle regimes of the stream. A domain-matched control that compares fresh and repeated data is the known follow-up.

We report no benchmark accuracy. We measure sink signatures with the probe of §2. We did not run downstream benchmark evaluation, such as MMStar, on any arm. We make no claim about how signature dissociation relates to downstream capability.

6. Conclusion

We tracked the three signatures that the field bundles as “the attention sink” as separate quantities across from-scratch multimodal pretraining. Those signatures are concentration, value-norm drain, and massive activation. They came apart everywhere we looked. Four training levers produced four different signature corners, and the value-norm axis alone moved in three different directions. On a repetition-confound-free, single-seed run over a billion fresh tokens, massive activation grew 130% while concentration never left zero. The per-head correlation between concentration and value-norm flipped sign across arms. The signatures even arrived in different orders. Norms came first and concentration never came in the softmax-from-scratch arms. Sigmoid attention mirrored that order. Text initialization delivered everything at once, inherited at step 0.

The signatures are plainly related in general. Work on text language models documents real interactions among them, which include a causal route from massive activations to sinks and compression valleys [2]. Our results show that the coupling is optional. In from-scratch multimodal pretraining, each axis moved independently under ordinary training-time levers. That extends the two-way text-only dissociations [3, 4] to a third axis and to a new setting. For interpretability work and mitigation work the practical result is blunt. One signature is not a proxy for the others. A model with no attention sink can still carry a growing massive activation. A gate that removes value-drain can leave everything else untouched.

Next steps. Four steps follow from this work. Train a randomly initialized vision encoder, to isolate the contribution of the decoder to massive activation. Run a per-position norm scan across all seeds, to close the remaining anchoring caveat on the text-initialized arm. Extend the fresh-data run past 1B tokens, to match text-LM budgets. Test whether the signature corner of an arm predicts hallucination or grounding behavior. Use the benchmarks that the sink-intervention literature already uses, such as POPE, CHAIR, and AMBER [21–23].

This paper does not test that last link. It establishes when the signatures form, and in what combinations, which is the measurement that has to exist first.

Reproducibility. A self-validating probe (§2) computes all signatures on a fixed probe batch, from dense logs taken every 100 steps. The Appendix holds the per-seed tables. We will release the code, the probe, the run configurations, the per-run logs, and the training checkpoints on acceptance. We withhold the links for double-blind review.

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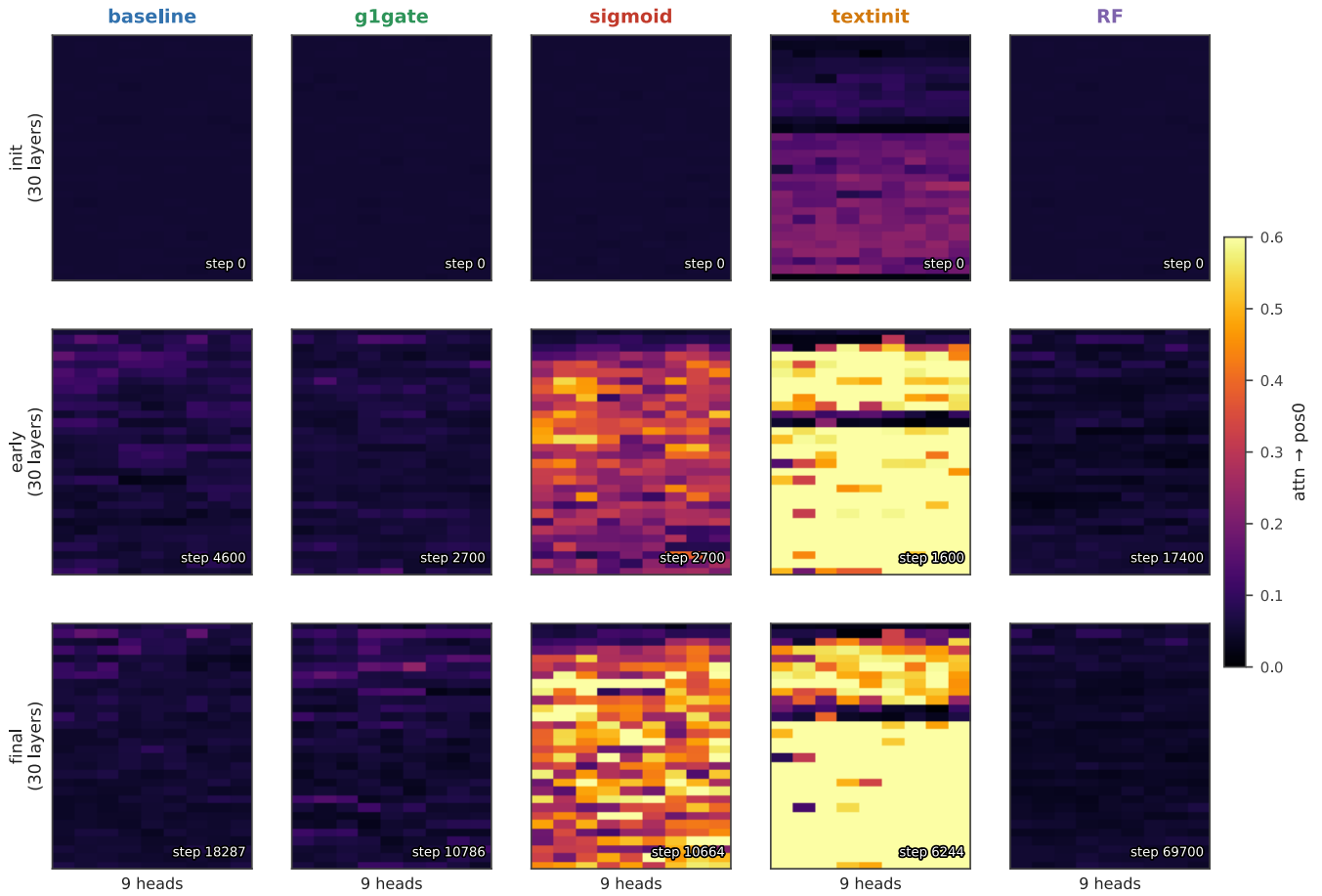
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Appendix

A. Supporting figures

Where the concentration sink lives: per-(layer,head) attn→pos0 over training



row-normalized attention (sigmoid raw gate ≈ 0.5 everywhere $\neq$ concentration); shared scale 0–0.6

Figure A1: Where concentration lives in the network. Attention to pos0 for each (layer, head) pair through training, row-normalized. baseline and RF stay cold throughout. textinit is hot from initialization. sigmoid lights up a band of mid-network layers.

No universal coupling: the concentration-value-norm sign is arm-dependent

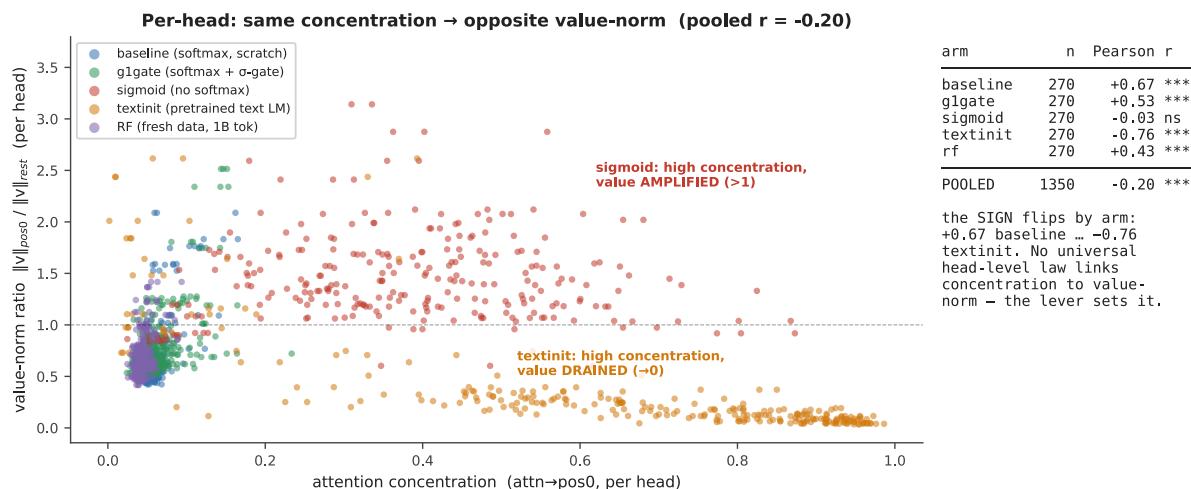


Figure A2: No universal head-level coupling. Concentration against value-norm ratio for each (layer, query-head) pair at the final checkpoint, by arm, at $n = 270$ per arm. Under grouped-query attention those pairs hold 90 independent value-norm observations (§3.3). The sign of the correlation flips by arm, from +0.67 in baseline to -0.76 in textinit (Table 4). The pooled cloud is weak, at -0.20, only because arms of opposite sign cancel. Do not read it as an uncorrelated cloud. Most individual arms show a strong $|r|$.

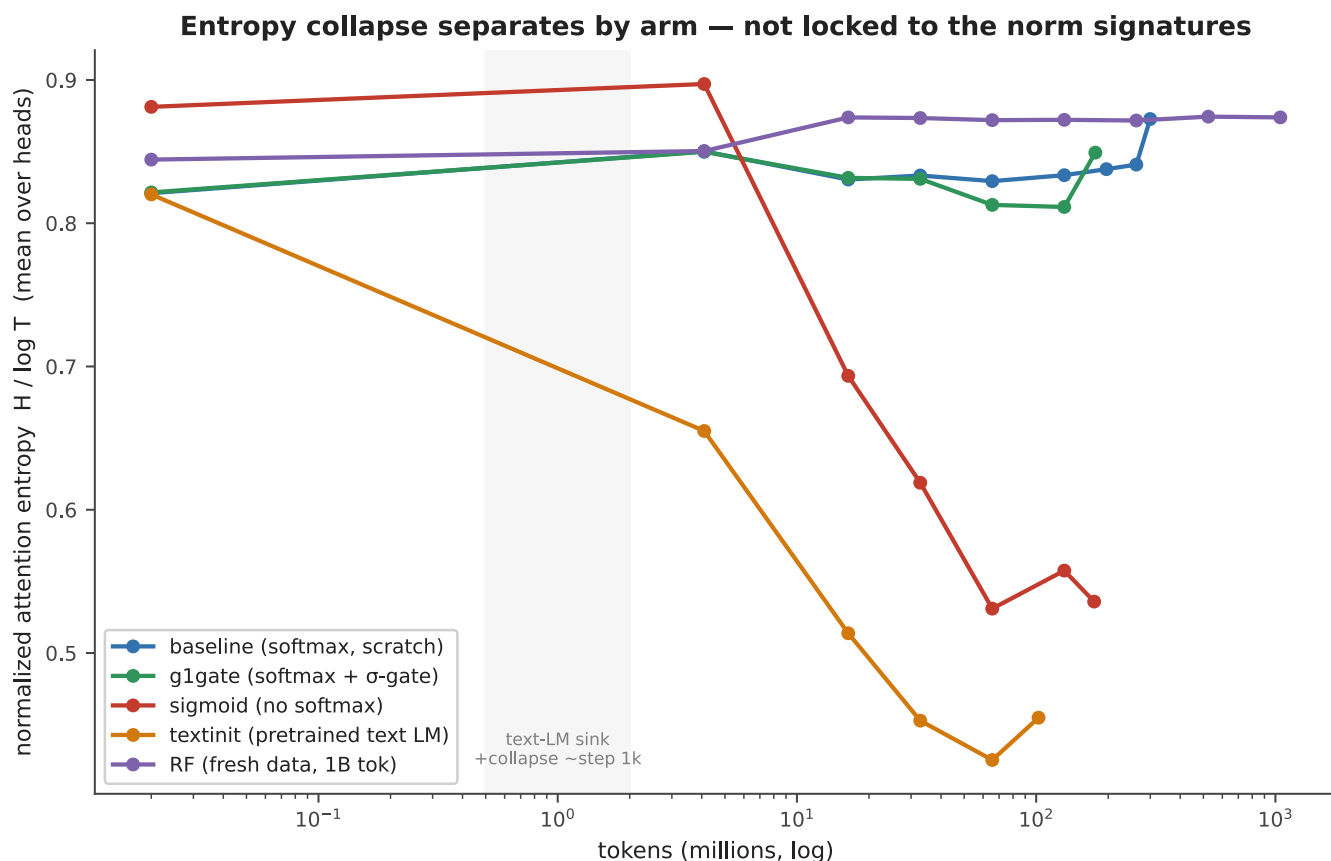


Figure A3: Entropy collapse tracks concentration only. Attention entropy through training. Only sigmoid and textinit collapse. baseline, glgate, and RF stay flat. The entropy-collapse correlate from the text-LM literature therefore follows the concentration axis, not the norm axes.

B. Full per-seed signature table

This table reproduces Table 1 and adds the maximum attention→pos0 where we have it. We logged that metric for seeds 1 and 2 only, because the seed-0 archive summary does not include it. We therefore keep it out of the main comparative table.

arm	seed	tokens	Sink ^{0.2}	mean attn→pos0	max attn→pos0	v-ratio	h-ratio
baseline	s0	101M	0.000	0.068	—	0.723	2.16
baseline	s1	100M	0.000	0.062	—	0.687	1.71
g1gate	s0	101M	0.004	0.068	—	0.805	1.67
g1gate	s1	100M	0.011	0.073	~0.21	0.845	2.04
g1gate	s2	100M	0.0037	0.072	~0.22	0.854	2.24
sigmoid	s0	102M	0.830	0.377	—	1.479	1.10
sigmoid	s1	100M	0.756	0.311	—	1.603	1.30
textinit	s0	60M	0.852	0.627	—	0.377	42.5
textinit	s1	60M	0.556	0.235	~0.53	0.634	5.50
textinit	s2	60M	0.578	0.232	~0.59	0.484	12.2

A note on g1gate. Both seeds with max-attention data show one head at about 0.21–0.22 maximum attention→pos0. The arm mean meanwhile stays near 0.07, and Sink^{0.2} stays far below 0.05. That pattern repeats across seeds, and it sits below every sink threshold this paper uses.

C. Per-position attention mass (seed 0)

This table gives the mean and max-head attention mass by position at seed 0. It confirms that position 0 is the maximum-mass token in every arm, by mass rather than by argmax count alone. It is the direct check behind the position-0 anchoring defense of §5.

arm	mass@pos0 (mean)	max-head@pos0	argmax-mass position	reading
baseline	0.06	0.17	0	pos0 max; diffuse (no sink)
g1gate	0.07	0.23	0	pos0 max; diffuse
sigmoid	0.30	0.66	0	pos0 max; broad raw-sigmoid mass
textinit	0.63	0.99	0	pos0 max; razor spike (pos1 = 0.009)

Per-position **mass** profiles for seeds 1 and 2 and for RF were not available when we wrote this paper. They need a checkpoint re-dump on a GPU. Only live-probe argmax data exists for those runs, and it is the source of the textinit position-migration caveat in §5.

D. Notes on metric hygiene

We record two corrections that we made during the audit of the numbers in this paper. First, an earlier internal consolidation mixed two metrics in one column, mean and max attention→pos0, and thereby manufactured an apparent seed anomaly in *g1gate*. Re-derived like-for-like, the anomaly disappears, and g1gate becomes the most reproducible arm. Second, an earlier draft claimed that the residual-norm peak of textinit “stays pinned

at pos0.” We checked that claim against the norm dump. It is wrong. The $\|h\|$ peak sits at pos1 or at pos13, depending on the seed. That is the positional-decoupling footnote of §3.4, which states the corrected form.